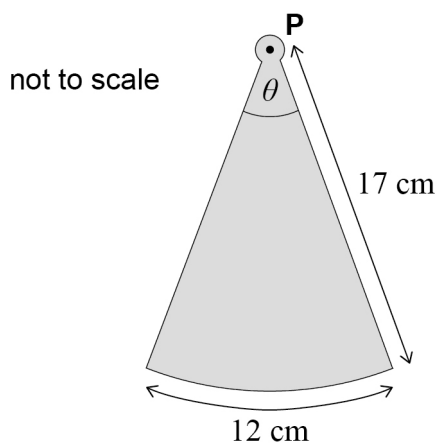


A thin aluminium sheet hangs from a pivot **P**. The sheet has an arc radius of 17 cm and an arc length of 12 cm, as shown in **Figure 1**.

Figure 1



The angle between the sides of the sheet is θ .

Show that θ is approximately 0.7 rad.

[1 mark]

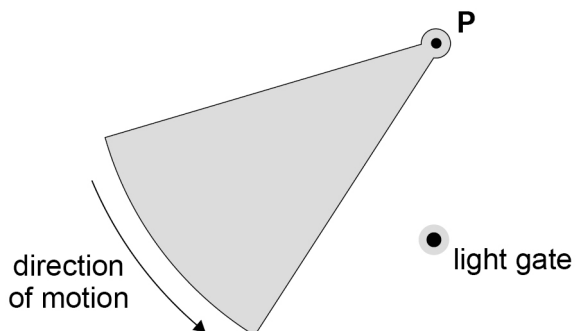
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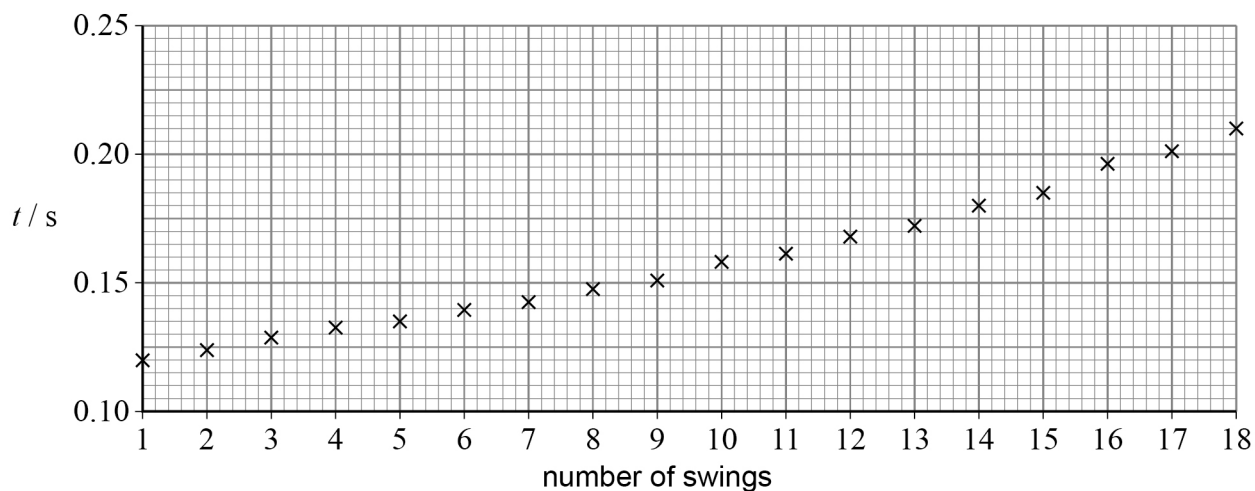


0 2 . 2

The sheet is pulled to one side and released to oscillate in a vertical plane. A light gate is placed vertically below **P**. The sheet swings from side to side, interrupting the light beam.

Figure 2

A datalogger records the time t for which the sheet interrupts the light beam during each swing. **Figure 3** shows the variation of t with the number of swings through the light gate.

Figure 3

The average angular speed during one swing of the sheet is $\bar{\omega}$.

Determine using **Figure 3** the largest value of $\bar{\omega}$.

[2 marks]

largest value of $\bar{\omega} =$ _____ rad s^{-1}

0 2 . 3

The sheet completes more than 18 swings before coming to a stop.

Results for t cannot be recorded after 18 swings.

Suggest why.

[1 mark]

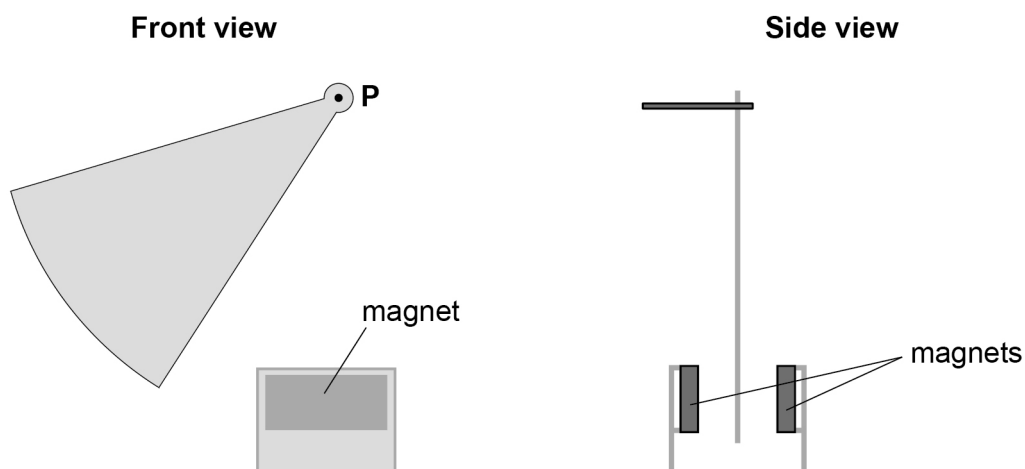
Question 2 continues on the next page



0 2 . 4

The aluminium sheet is now made to swing through a uniform magnetic field between two magnets as shown in **Figure 4**.

Figure 4



The sheet is released from the same initial position as in question **02.2**.

The presence of the magnetic field increases the damping of the oscillations of the sheet.

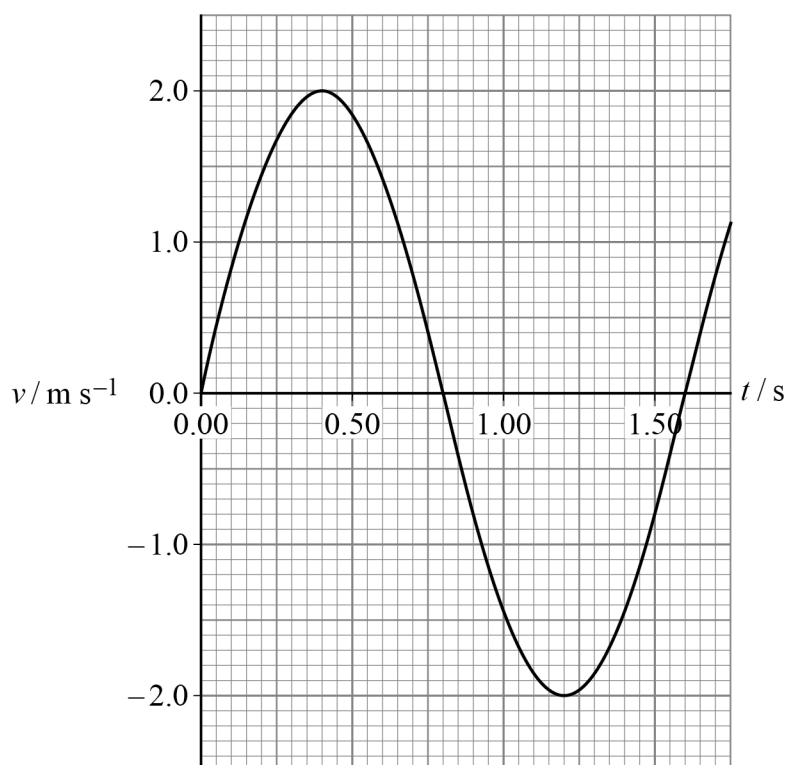
Explain why.

[3 marks]



Do not write

The graph shows the variation of velocity v with time t for an object undergoing simple harmonic motion.



What is the amplitude of oscillation?

[1 mark]

A 1.40 m

☐

B 0.80 m

☐

C 0.54 m

☐

D 0.51 m

☐

Turn over for the next question

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Do not write Do not write

1

1

A mass–spring system of mass m oscillates with frequency f on a spring with spring constant k .

A second mass–spring system oscillates at frequency $3f$.

Which row is correct for the second system?

[1 mark]

	Mass	Spring constant	
A	$18m$	$2k$	☐
B	$9m$	$3k$	☐
C	$3m$	$9k$	☐
D	$2m$	$18k$	☐

1**2**

The variation of displacement x with time t for an oscillator of mass m is

$$x = 2.4\cos(0.76t)$$

What is the total energy of the oscillator?

[1 mark]

A $0.9m$

☐

B $1.4m$

☐

C $1.7m$

☐

D $3.3m$

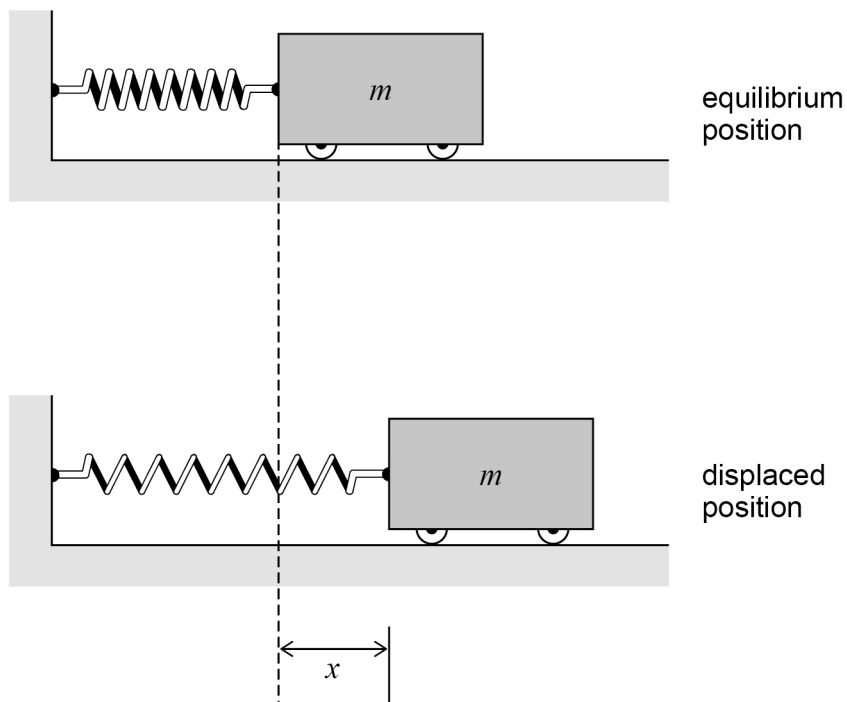
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Don't write

01

A student connects a trolley of mass m to a horizontal spring, as shown in **Figure 1**. The trolley is displaced horizontally from the equilibrium position and oscillates. Any damping of the trolley's motion is negligible.

Figure 1



01.1

The student uses a stopwatch to determine the time period T of the mass-spring system.

State **two** procedures the student can use to minimise the uncertainty in T .

[2 marks]

1 _____

2 _____



0 1 . 2

The resultant force F on the oscillating trolley varies with its displacement x from the equilibrium position by:

$$F = -kx$$

where k is the spring constant.

Explain the physical significance of the negative sign in this equation.

[1 mark]

0 1 . 3

Show how the equation for the resultant force F on the trolley can be combined with $F = ma$ and other equations to give

$$T = 2\pi\sqrt{\frac{m}{k}}$$

[4 marks]

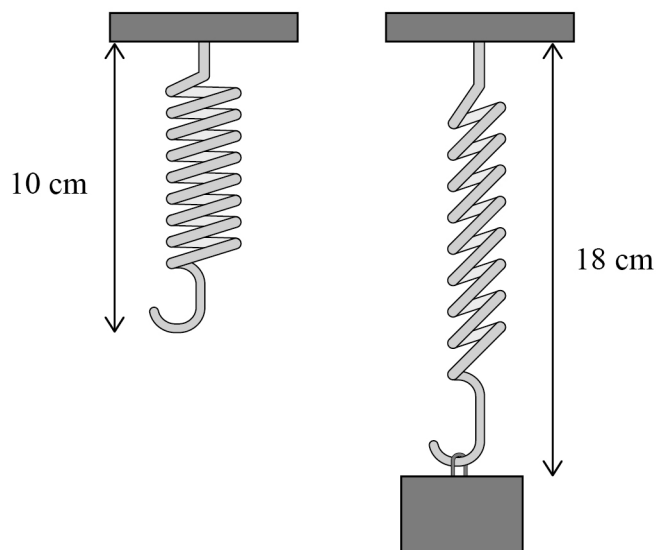
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Do not write

The unextended length of a light spring is 10 cm.

The length of the spring is 18 cm when a 400 g mass is suspended from it.



The mass–spring system oscillates with a small amplitude.

What is the period of the oscillation?

[1 mark]

A 0.57 s

☐

B 0.85 s

☐

C 5.7 s

☐

D 8.5 s

☐

Turn over for the next question



Do not write

An equation describing a system undergoing simple harmonic motion is

$$v = \omega \sqrt{(A^2 - x^2)}$$

What is x ?

[1 mark]

A $x = \omega A - v$

☐

B $x = \sqrt{\omega A - v}$

☐

C $x = A - \frac{v}{\omega}$

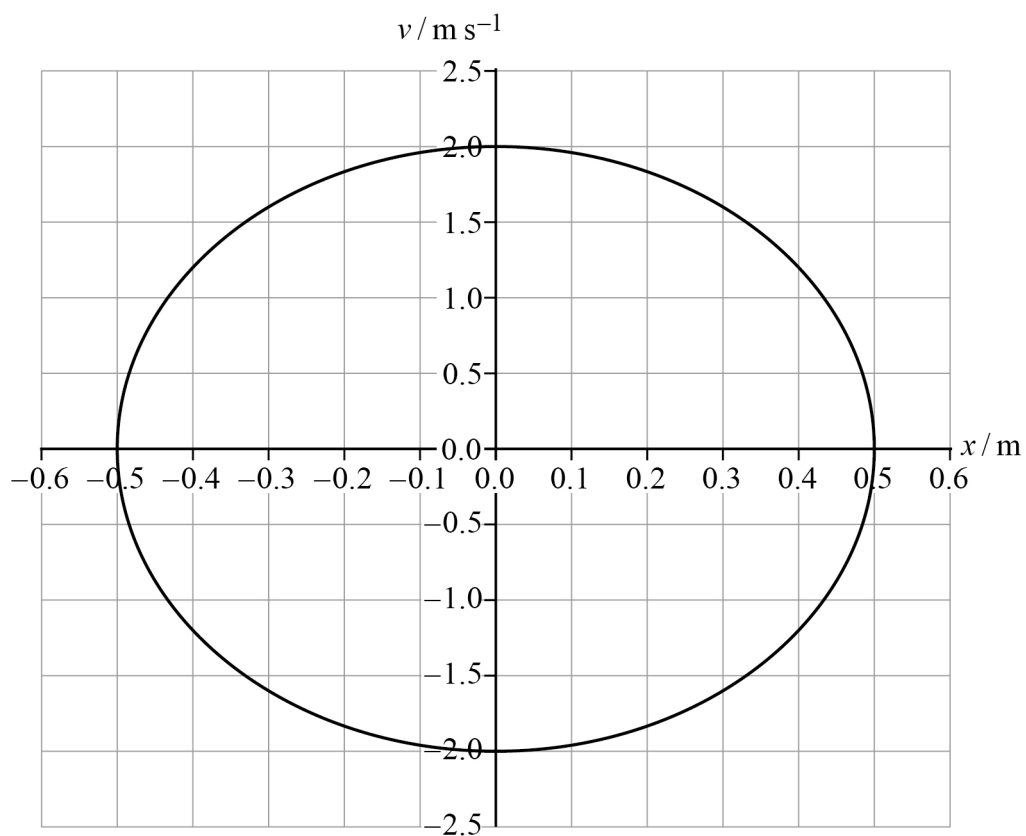
☐

D $x = \sqrt{A^2 - \frac{v^2}{\omega^2}}$

☐

Don't waste

The graph shows the variation of velocity v with displacement x for a simple harmonic oscillator.



What is the maximum acceleration of the oscillator?

[1 mark]

A 1.0 m s^{-2}

☐

B 4.0 m s^{-2}

☐

C 8.0 m s^{-2}

☐

D 16 m s^{-2}

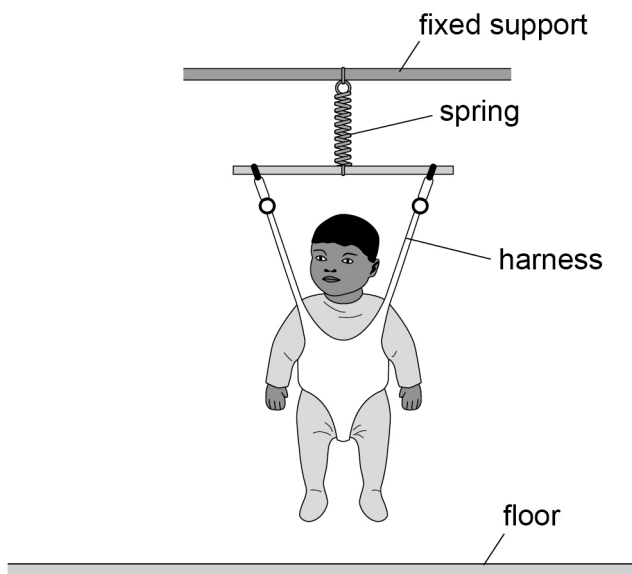
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Don't write

0 2 **Figure 1** shows a mass–spring system that consists of a baby in a harness. The system is suspended from a fixed support by a vertical spring.

The total mass of the baby and harness is 8.5 kg.
The spring has negligible mass and obeys Hooke's Law.

Figure 1



0 2 . 1 The extension of the spring is 0.16 m when the system is in equilibrium.

Show that the spring constant of the spring is approximately 520 N m^{-1} .

[1 mark]



The baby is displaced vertically downwards by 4.0 cm from the equilibrium position and released at time $t = 0$

The system then oscillates vertically with simple harmonic motion.

0 2 . 2

Show that the time period of the oscillation is approximately 0.8 s.

[1 mark]

0 2 . 3

Calculate the initial total energy of the system due to the oscillation.

[3 marks]

initial total energy = _____ J

Question 2 continues on the next page

Turn over ►



The baby does not add energy to the oscillating system.

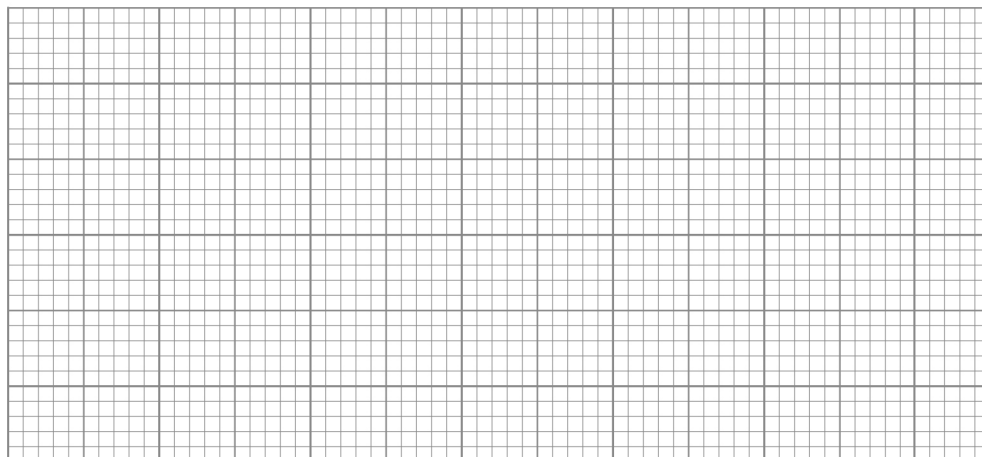
0 2 . 4

Sketch a graph on **Figure 2** to show the variation of the kinetic energy of the system with time from $t = 0$ to $t = 1.2$ s.

Label the axes and add appropriate scales.

[4 marks]

Figure 2



0 2 . 5

In practice, the oscillations are damped by air resistance.

Describe how the damping force varies with time.

[2 marks]



Don't write

An object is performing simple harmonic motion. There are phase differences between the acceleration, displacement and velocity of the object.

Which row is correct?

[1 mark]

	Phase difference between the acceleration and the displacement / rad	Phase difference between the acceleration and the velocity / rad	
A	0	π	☐
B	π	$\frac{\pi}{2}$	☐
C	0	$\frac{\pi}{2}$	☐
D	π	π	☐



do not write

2

1

A mains power supply has a sinusoidal voltage with a root mean square value of 230 V.

What is the peak-to-peak voltage?

[1 mark]

A 163 V

☐

B 325 V

☐

C 460 V

☐

D 651 V

☐

[illegible]

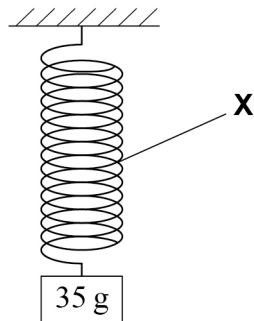
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Figure 1 shows a load of mass 35 g attached to a vertical spring **X**. The load is at rest in its equilibrium position.

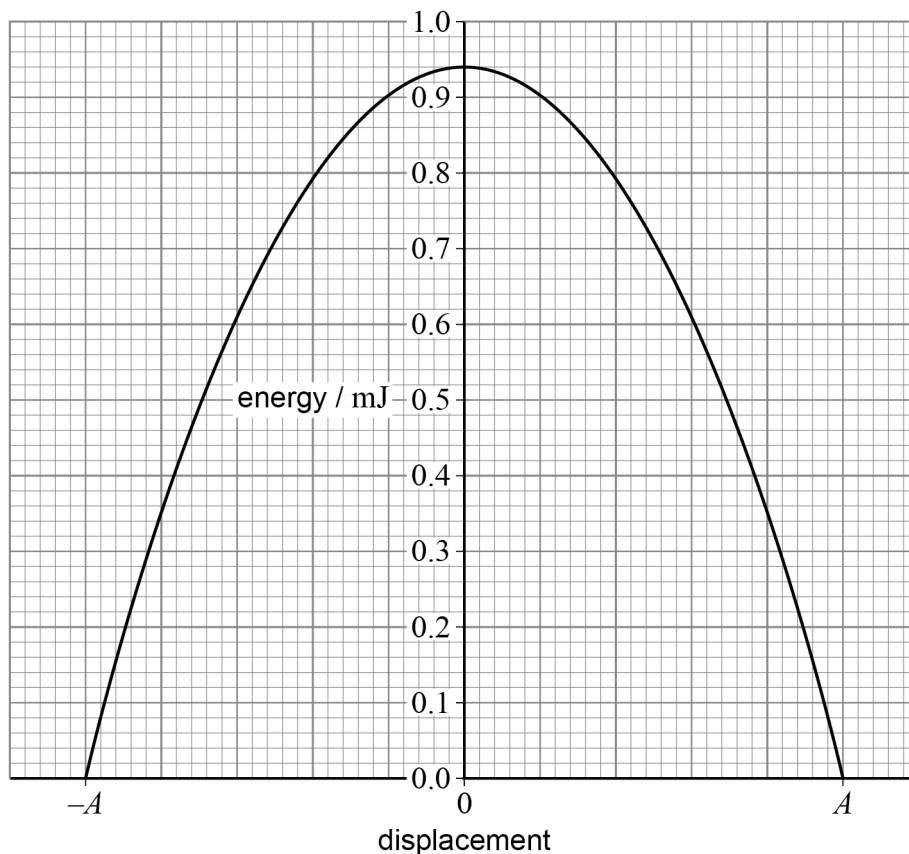
Figure 1



The load is pulled vertically downwards below its equilibrium position and released. The load oscillates with simple harmonic motion (SHM) with a period of 0.68 s and amplitude A .

Figure 2 shows the variation of kinetic energy of the load with its displacement from the equilibrium position during an oscillation.

Figure 2



0 1 . 1 Calculate, in mm, A .

[4 marks]

$A =$ _____ mm

0 1 . 2

The load is brought to rest and then pulled vertically downwards a distance $\frac{A}{2}$ below its equilibrium position.

It is released and oscillates with SHM.

Spring **X** has negligible mass.

Sketch, on **Figure 2**, the variation of the **total** energy of the mass–spring system with displacement.

[2 marks]

Question 1 continues on the next page

Turn over ►



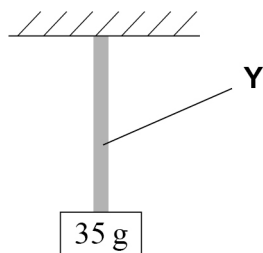
Spring **X** is replaced with an elastic string **Y** that obeys Hooke's law.

Figure 3 shows the load attached to the end of **Y**.

Y has negligible mass.

Y has an extension of 42 mm when the load is at rest in its equilibrium position.

Figure 3



The load is pulled 20 mm vertically downwards below its equilibrium position. It is then released and oscillates with SHM.

Show that the period of the oscillations is approximately 0.4 s.

[2 marks]

The load on **Y** is now pulled 50 mm vertically downwards below its equilibrium position and released.

The movement of the load fulfils the conditions for SHM until the load reaches a position **P**. When the load is at **P**, the elastic string **Y** is at its unstretched length.

Calculate the velocity of the load at **P**.

[3 marks]

velocity = _____ m s^{-1}



0 1 . 5

The load continues to move upwards after reaching **P**.

Explain why this movement above **P** does **not** fulfil the conditions for SHM.

[2 marks]

0 1 . 6

State whether spring **X** or elastic string **Y** has the greater stiffness.
Explain your answer.

[2 marks]

15

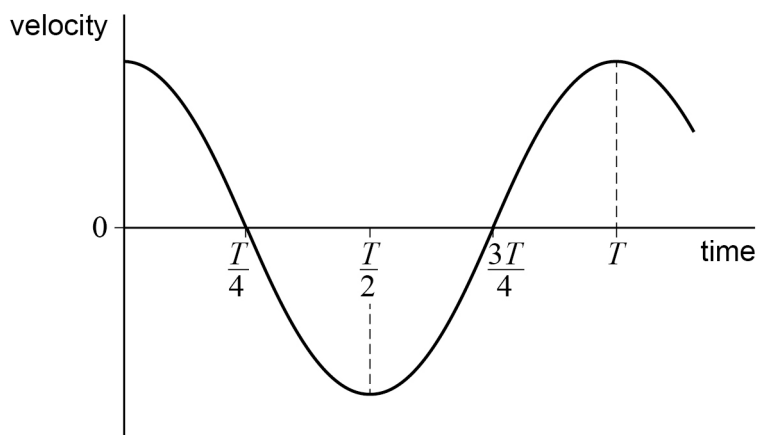
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Do not write Do not write

The graph shows the variation with time of the velocity of a particle performing simple harmonic motion.



Which statement about the particle is correct?

[1 mark]

A At time $\frac{T}{4}$ the restoring force on the particle is zero.

☐

B At time $\frac{T}{2}$ the potential energy E_p of the particle is a minimum.

☐

C At time $\frac{3T}{4}$ the displacement of the particle is zero.

☐

D At time T the acceleration of the particle is a maximum.

☐

Turn over for the next question

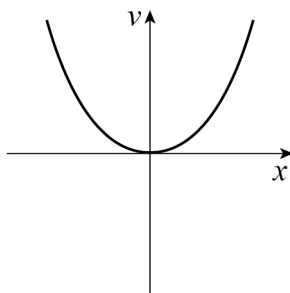
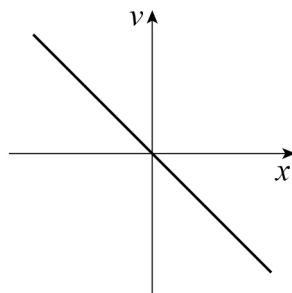
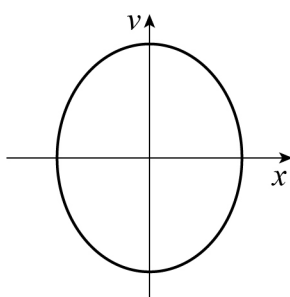
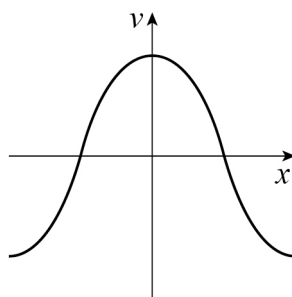


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0 8

A body is moving with simple harmonic motion.

What is the variation of its velocity v with its displacement x ?

[1 mark]**A****B****C****D****A**☐**B**☐**C**☐**D**☐

Do not write Do not write

0 9

A mass oscillates with amplitude A and frequency f on a spring of spring constant k . The total energy of the mass–spring system is E .

A second mass–spring system has an identical mass and a different spring. The second system oscillates with amplitude $2A$ and frequency $2f$.

Which shows the spring constant of the second spring and the total energy of the second system?

[1 mark]

	Spring constant	Total energy	
A	$\sqrt{2}k$	$4E$	☐
B	$2k$	$16E$	☐
C	$4k$	$4E$	☐
D	$4k$	$16E$	☐

A pendulum bob is pulled to one side so that its string makes an angle θ to the vertical. The bob is released and the pendulum oscillates freely.

What is the maximum acceleration of the bob?

[1 mark]

A g

☐

B $g \cos \theta$

☐

C $g \sin \theta$

☐

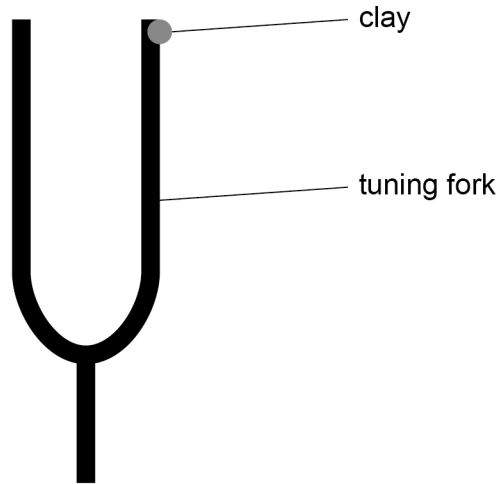
D $g \tan \theta$

☐

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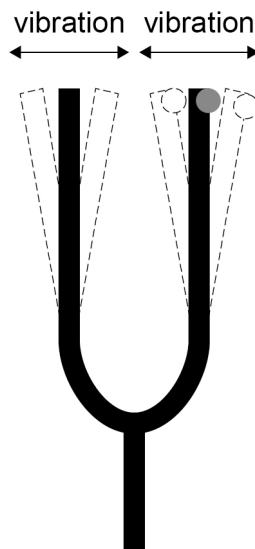
Figure 2 shows a tuning fork with a small piece of clay attached to it.

Figure 2



The tuning fork is made to vibrate. The clay stays attached to the tuning fork.
Figure 3 shows the tuning fork and clay vibrating.

Figure 3

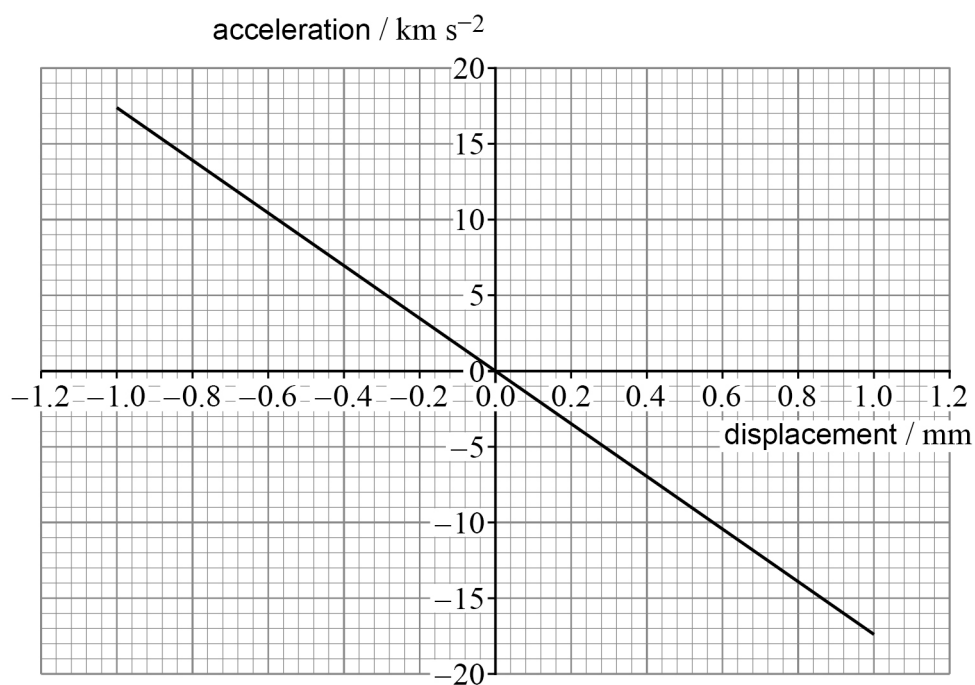


Question 2 continues on the next page



Figure 4 shows the variation of acceleration of the clay with displacement from its equilibrium position.

Figure 4



0 2 . 1

Explain how **Figure 4** shows that the motion of the clay is consistent with the conditions for simple harmonic motion.

[2 marks]



0 2 . 2 Show that the frequency of the vibration of the clay is approximately 700 Hz.

[3 marks]

0 2 . 3 Calculate, in m s^{-1} , the maximum speed of the clay.

[2 marks]

maximum speed = _____ m s^{-1}

0 2 . 4 Label, with a **P** on **Figure 4**, the position of a point at which the clay is travelling at its maximum speed.

[1 mark]



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ANSWER IN THE SPACES PROVIDED**



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A particle undergoes SHM with a period of 4.6 s.

The particle has a speed of 1.25 m s^{-1} when its displacement from the centre is 0.60 m.

What is the amplitude of the particle's oscillation?

[1 mark]

A 0.72 m

☐

B 0.84 m

☐

C 1.09 m

☐

D 1.20 m

☐

do not write

A particle experiences SHM.
Its potential energy is E when at its maximum displacement.

Which row identifies the total energy and kinetic energy of the particle when its velocity is a maximum?

[1 mark]

	Total energy	Kinetic energy	
A	E	$\frac{E}{2}$	☐
B	E	E	☐
C	$2E$	E	☐
D	$2E$	$2E$	☐

2	1
---	---

An object with a mass of 0.15 kg is suspended from a spring.
The spring has a stiffness of 2.5 N m^{-1} .
The object is displaced vertically and released.
It undergoes SHM with a period T .

Which length of simple pendulum also has a period T ?

[1 mark]

A 0.4 m

☐

B 0.6 m

☐

C 3.7 m

☐

D 5.8 m

☐



2 2

An object is suspended from a spring of stiffness k .
The extension of the spring is ΔL .
The object is displaced vertically and released.
It oscillates with SHM of frequency f .

The procedure is repeated using the same object and a spring with stiffness $2k$. The values of ΔL and f are different for this new spring.

Which quantity doubles when the second spring of stiffness $2k$ is used?

[1 mark]

A $\left(\frac{1}{f\Delta L}\right)^2$ ☐

B $\frac{1}{f\Delta L}$ ☐

C $f\Delta L$ ☐

D $(f\Delta L)^2$ ☐

[illegible]

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Figure 4 shows a pendulum of length 1.80 m. A rod is a distance h vertically below the point where the pendulum string is clamped.

Figure 4

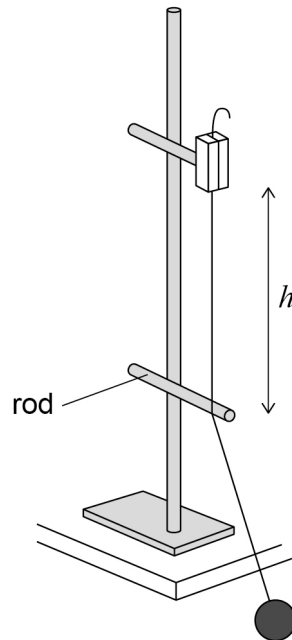
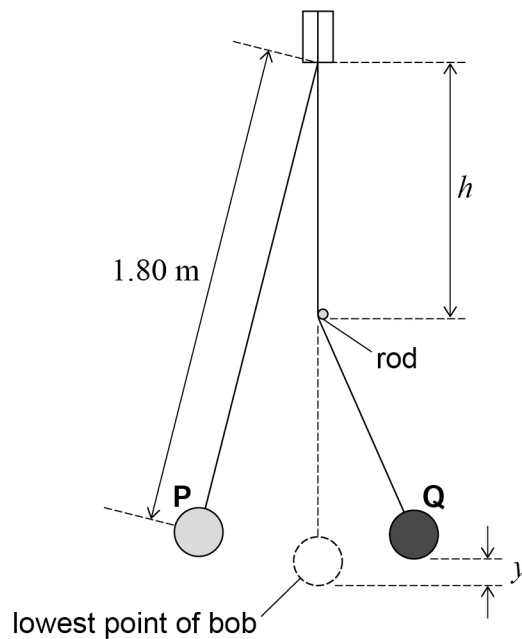


Figure 5 shows a view of the pendulum from the front.

Figure 5



not to scale

P and **Q** are the positions of the bob when it is at its maximum displacement on the left-hand side and the right-hand side respectively.
The vertical distance of the bottom of the bob above its lowest point is y .
In **Figure 5**, y is shown at a maximum value.

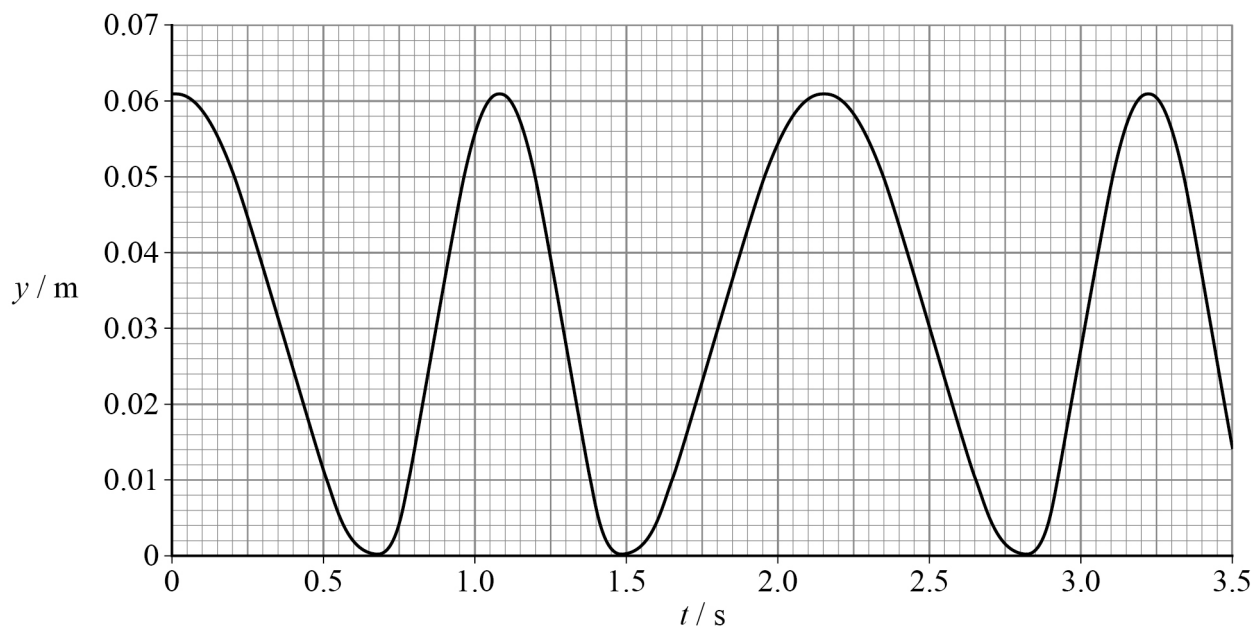


The pendulum is released from **P** at time $t = 0$

The string makes contact with the rod when the bob is at the lowest point of its oscillation. At this point, only the lower part of the pendulum continues to oscillate. The bob comes to rest at **Q** for an instant. The bob continues to move between **Q** and **P**.

Figure 6 shows the variation of y with time t .

Figure 6



For the time shown in **Figure 6** the effect of air resistance is negligible.

0 2 . 1

Explain how **Figure 6** shows that no energy is transferred from the system.

[2 marks]

Question 2 continues on the next page

Turn over ►



0 2 . 2 Determine the speed of the bob when $y = 0$

[3 marks]

speed = _____ m s^{-1}

0 2 . 3 Determine h .
Use data from **Figure 5** and **Figure 6**.

[4 marks]

$h =$ _____ m

Turn over for the next question

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ANSWER IN THE SPACES PROVIDED**

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Do not write

A mass-spring system is oscillating with simple harmonic motion (SHM).

The kinetic energy E_k and the displacement x of the mass vary between zero and maximum values.

Which row describes E_k and x at the instant when the momentum of the mass is at a maximum?

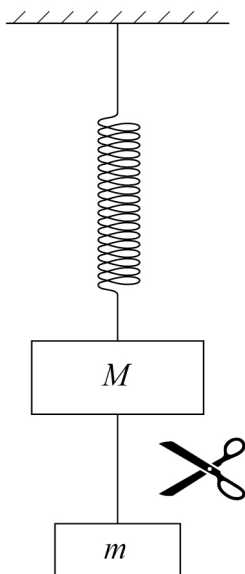
[1 mark]

	E_k	x	
A	maximum	zero	☐
B	maximum	maximum	☐
C	zero	zero	☐
D	zero	maximum	☐



Don't write

A small object of mass m is suspended from a larger object of mass M using a thread of negligible mass. The system is in equilibrium and the total extension of the spring is Δl .



The thread is cut and the object of mass M moves with simple harmonic motion (SHM).

What is the time period of the SHM?

[1 mark]

A $2\pi\sqrt{\frac{M\Delta l}{(M-m)g}}$ ☐

B $2\pi\sqrt{\frac{M\Delta l}{mg}}$ ☐

C $2\pi\sqrt{\frac{M\Delta l}{(M+m)g}}$ ☐

D $2\pi\sqrt{\frac{(M+m)\Delta l}{Mg}}$ ☐

Turn over for the next question



Do not write

2 **1** An object is moving with simple harmonic motion (SHM) with a period T .
The total energy of the system is E .

The object is at its maximum displacement at time $t = 0$

Which row gives the kinetic energy E_k of the object when $t = \frac{T}{4}$ and when $t = \frac{T}{2}$?

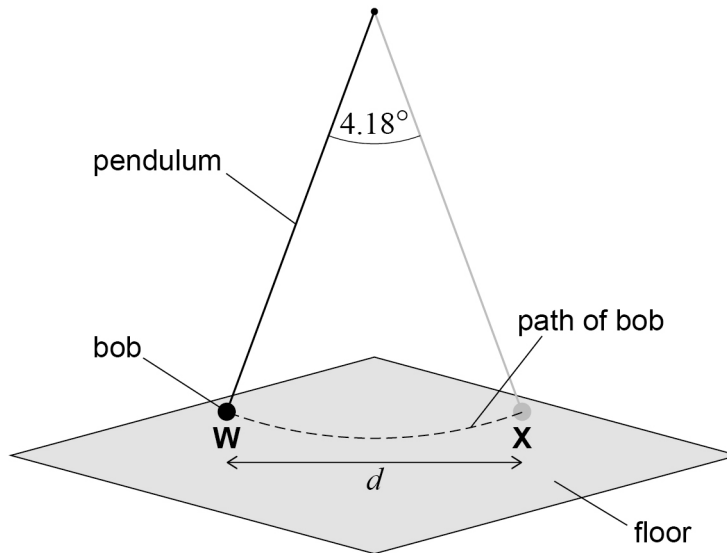
[1 mark]

	E_k when $t = \frac{T}{4}$	E_k when $t = \frac{T}{2}$	
A	E	0	☐
B	0	E	☐
C	0	$\frac{E}{4}$	☐
D	E	$\frac{3E}{4}$	☐

Figure 1 shows a simple pendulum of length 67.7 m.
The pendulum bob is released from position **W** and swings in a vertical plane.

Figure 1

not to scale



The pendulum swings through an angle of 4.18° . The bob just reaches position **X**.
The horizontal distance between **W** and **X** is d .

Air resistance is negligible.

Calculate d .

[2 marks]

$d =$ _____ m



0 1 . 2 Calculate the maximum speed of the bob.

[3 marks]

maximum speed = _____ m s^{-1}

Question 1 continues on the next page

Turn over ►

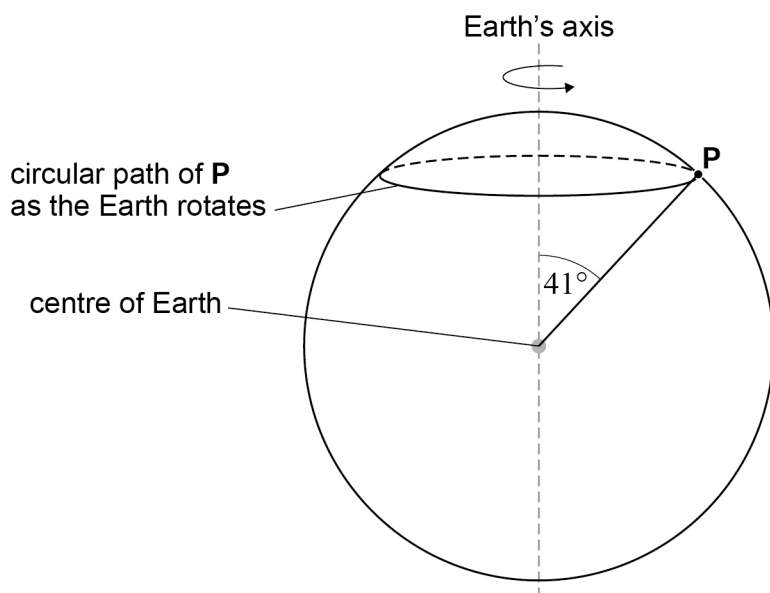


The pendulum is at point **P** on the surface of the Earth.

Figure 2 shows **P**.

The line joining **P** to the centre of the Earth is at an angle of 41° to the Earth's axis.
Due to the Earth's rotation, **P** travels in a circular path about the Earth's axis.

Figure 2



Calculate the linear speed of **P** due to the rotation of the Earth.

[2 marks]

speed = _____ m s^{-1}



0 1 . 4

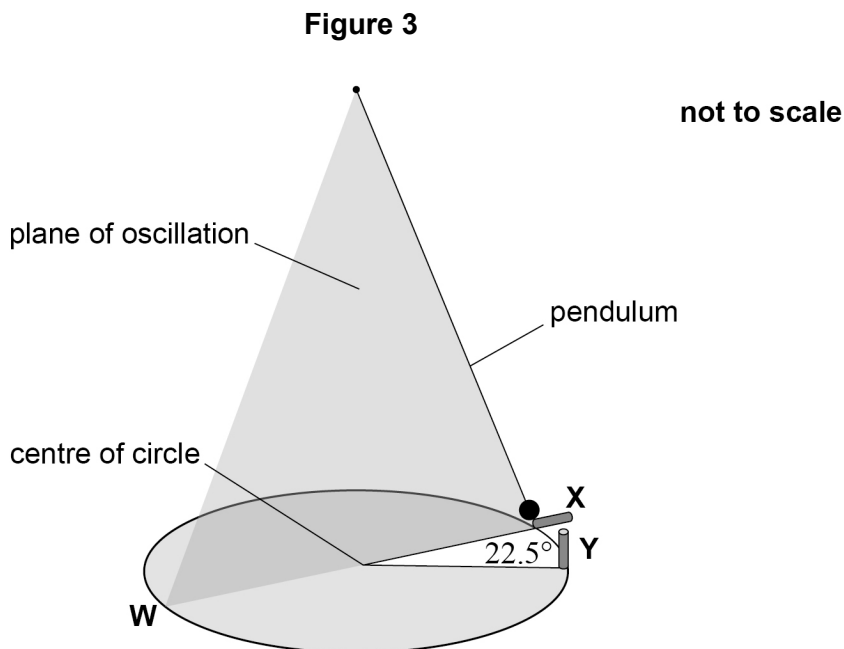
Due to the rotation of the Earth, the plane of oscillation of the pendulum rotates slowly about a vertical axis relative to the floor.

The plane of oscillation rotates through 360° in 31.8 hours.

Pegs are placed on the floor at positions **X** and **Y** on a circle.

The angle between **X** and **Y**, measured at the centre of the circle, is 22.5° as shown in **Figure 3**.

Figure 3 shows the pendulum just after it knocks over the peg at **X**.



The plane of oscillation continues to rotate relative to the floor. The amplitude of oscillation of the pendulum is unchanged.

After knocking over the peg at **X**, the pendulum goes through n oscillations before knocking over the peg at **Y**.

Calculate n .

[3 marks]

$n =$ _____

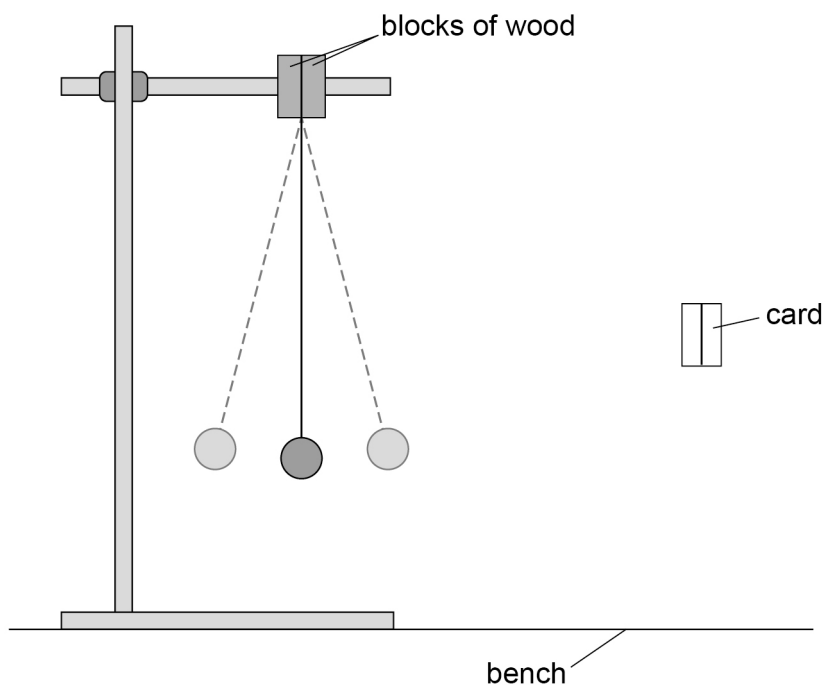
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07

Figure 11 shows a simple pendulum used in an experiment to measure the acceleration g due to gravity.

Figure 11 also shows a card marked with a vertical line.

Figure 11



07.1

Annotate **Figure 11** with a double-headed arrow to show the length L of the pendulum.

[1 mark]

07.2

The card is to be used as a fiducial marker to reduce the uncertainty in the measurement of the period T of the pendulum.

Draw the card in a suitable position on **Figure 11**.

Go on to explain why you have chosen this position.

[2 marks]

Question 7 continues on the next page

Turn over ►



0 7 . 3

Describe **two** other techniques that can be used to minimise the uncertainty in T .

[2 marks]

1 _____

2 _____

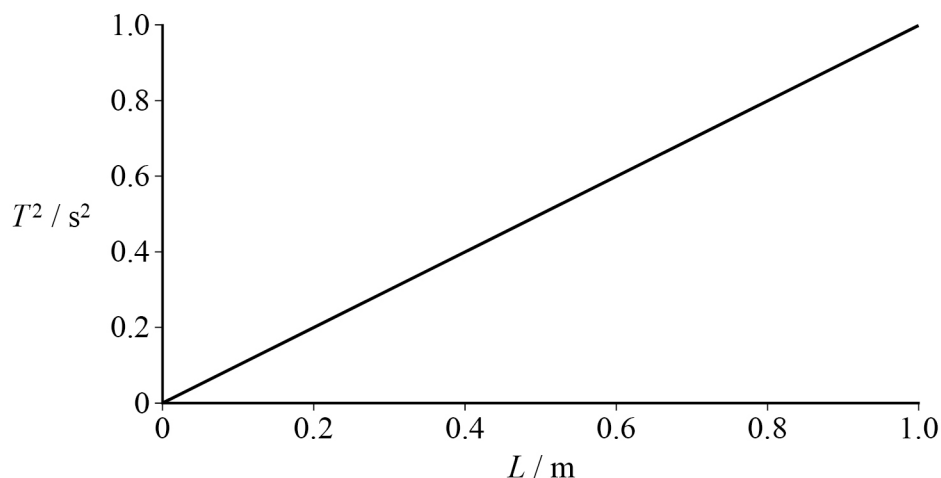


07.4

A student varies L and attempts to measure the period T for each value of L .

Figure 12 shows the student's graph of their T^2 against L .

Figure 12



Due to a mistake in their measurement of T , the student obtains a value for g of 39.1 m s^{-2} .

Suggest the mistake made by the student.

Go on to determine the correct value for g for the student's data.

[2 marks]

$g =$ _____ m s^{-2}

7

END OF SECTION A

Turn over ►



Section B

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each question, completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown.



If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown.



You may do your working in the blank space around each question but this will not be marked.
Do **not** use additional pages for this working.

An object moves with simple harmonic motion of amplitude 0.80 m and period 9.0 s.

What is the speed of the object when its displacement is 0.60 m?

[1 mark]

A 0.14 m s⁻¹

☐

B 0.20 m s⁻¹

☐

C 0.37 m s⁻¹

☐

D 0.42 m s⁻¹

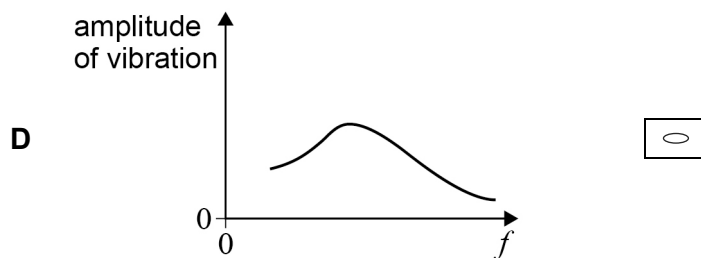
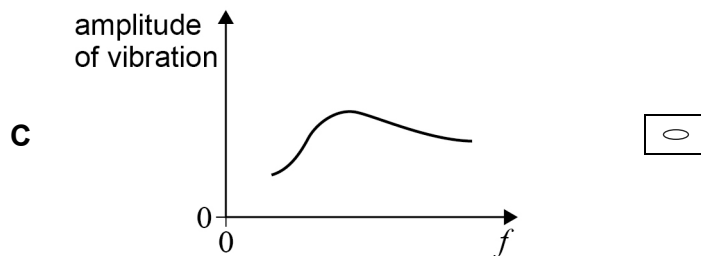
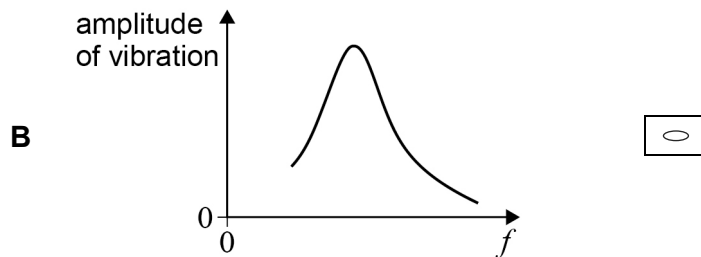
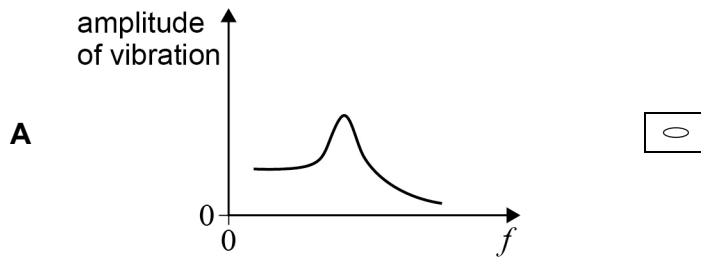
☐

Do not write

A damped oscillator is set into forced vibrations by a periodic driving force of frequency f . Graphs **A**, **B**, **C** and **D** are drawn to the same scale.

Which graph shows the heaviest damping?

[1 mark]



Do not write

2**2**

A sinusoidal signal with an rms voltage of 80 V is connected across the inputs of an oscilloscope.

The y -gain of the oscilloscope is 40 V cm^{-1} .

What is the amplitude of the sinusoidal waveform displayed on the screen?

[1 mark]

A 1.4 cm

☐

B 2.0 cm

☐

C 2.8 cm

☐

D 5.6 cm

☐

[illegible]



[illegible]

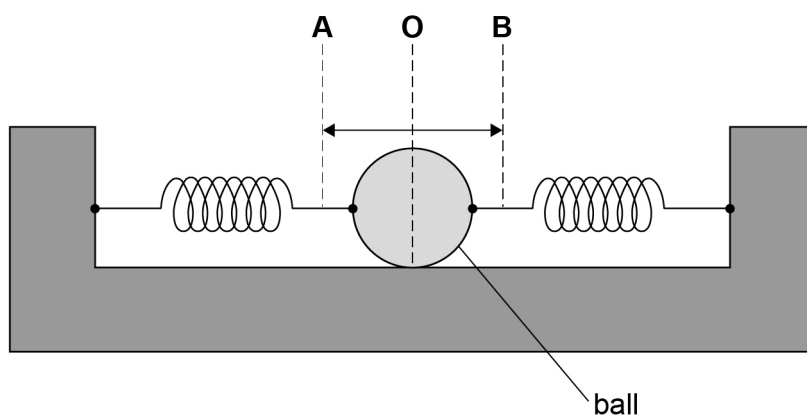
06

This question is about the simple harmonic motion of a mass–spring system.

Figure 10 shows a ball held in position by springs.

The equilibrium position of the ball is at **O**.

Figure 10



The centre of the ball is displaced to **B**, a distance of 3.6 cm to the right of **O**.

The ball is released at time $t = 0$ and then oscillates horizontally between **A** and **B**.

The frequency of the oscillation is 1.7 Hz.

06.1

Calculate the minimum time taken for the centre of the ball to travel from **B** to **O**.

[1 mark]

minimum time = _____ s



0 6 . 2

Calculate the minimum time taken for the centre of the ball to travel from **B** to a point 1.8 cm to the left of **O**.

[2 marks]

minimum time = _____ s

0 6 . 3

The maximum kinetic energy of the ball is 3.8 mJ.

Calculate the mass of the ball.

[2 marks]

mass = _____ kg

Question 6 continues on the next page

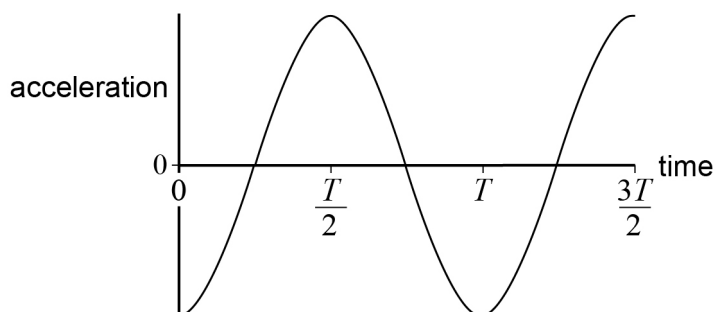
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0 6 . 4

Figure 11 shows the variation of acceleration with time for the ball.
 T is the period of the oscillation.

Figure 11

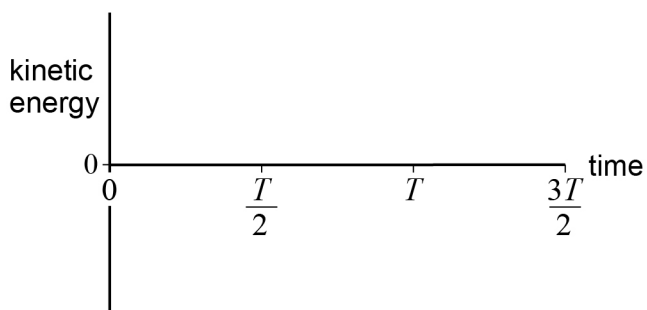


Sketch, on **Figure 12**, the variation of the kinetic energy of the ball with time, for the time shown in **Figure 11**.

Do not include numerical values on the axes.

[2 marks]

Figure 12



Turn over for the next question

*Do not write
outside the
box*

**DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED**

Turn over ►



Do not write

1

1

outside the
box

A particle undergoes simple harmonic motion.

What is the phase difference between the particle's displacement and its acceleration?

[1 mark]

A 0



B $\frac{\pi}{4}$ rad



C $\frac{\pi}{2}$ rad

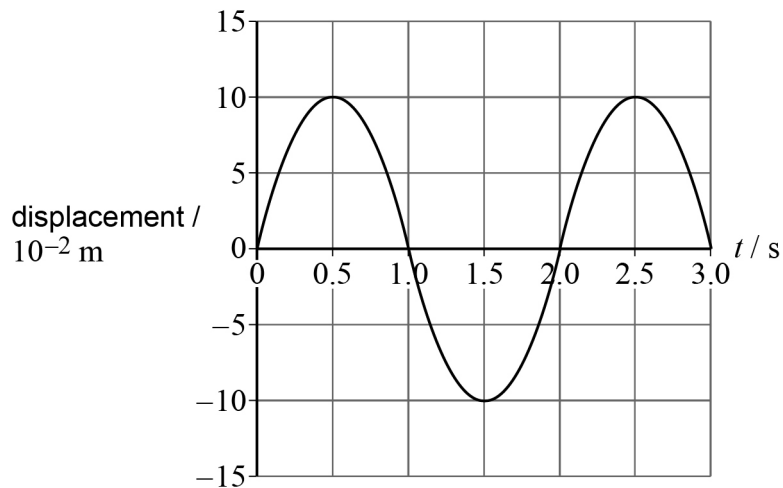


D π rad



1 2

An object of mass 0.45 kg undergoes simple harmonic motion.
The graph shows the variation of the object's displacement with time t .



What is the potential energy of the object when $t = 1.5 \text{ s}$?

[1 mark]

A 0

☐

B $1.1 \times 10^{-2} \text{ J}$

☐

C $2.2 \times 10^{-2} \text{ J}$

☐

D $8.9 \times 10^{-2} \text{ J}$

☐

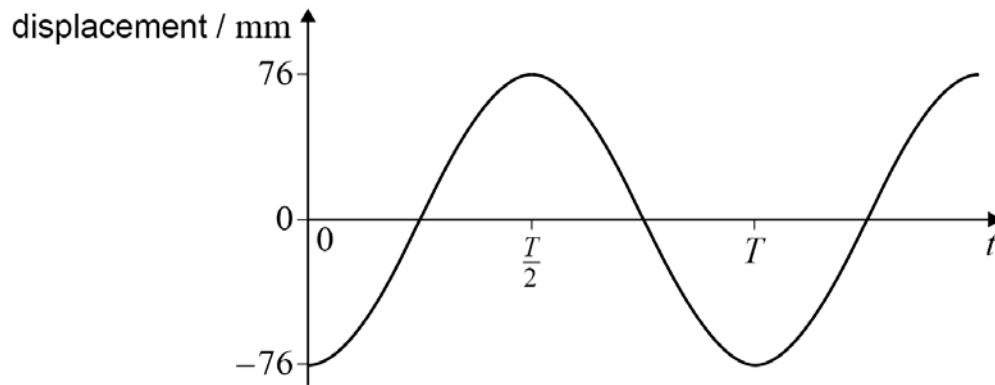
Turn over ►



Do not write

A 200 g mass is suspended from a vertical spring. Initially, the mass–spring system is at rest. The mass is pulled 76 mm below its rest position and released. **Figure W** shows the displacement of the mass from its rest position with time t . The time period for one oscillation is T .

Figure W



State **two** conditions that must apply to the acceleration of the mass.

[2 marks]

condition 1 _____

condition 2 _____

Question 6 continues on the next page

0 6 . 2 The time taken for 25 oscillations is 23 s.

Show that the frequency of the oscillation is approximately 1.1 Hz.

[1 mark]

0 6 . 3 Calculate the maximum acceleration of the mass.

[2 marks]

maximum acceleration = _____ m s⁻²

0 6 . 4 Calculate the magnitude of the displacement of the mass from its rest position 0.60 s after being released.

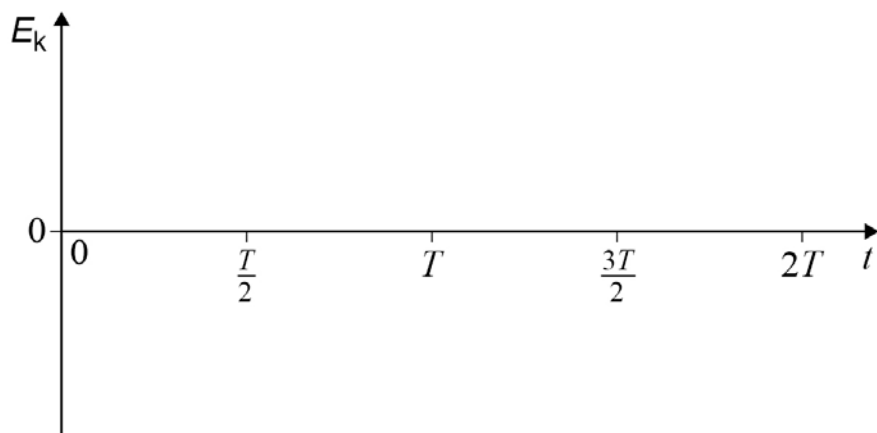
[2 marks]

displacement = _____ mm

0 6

. 5

Sketch, on the axes in **Figure X**, the variation of kinetic energy E_k of the mass with time t for **two** complete oscillations of the mass–spring system. The mass is released from 76 mm below its rest position at $t = 0$

[2 marks]**Figure X**

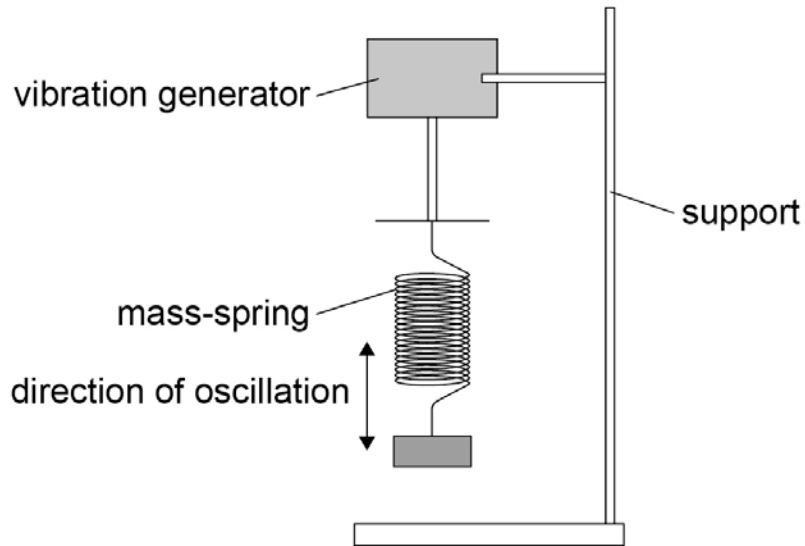
Question 6 continues on the next page

0 6

6

Figure Y shows a vibration generator attached to the mass-spring system. The vibration generator vibrates with a small amplitude at a frequency of 1.1 Hz.

Figure Y



The small-amplitude vibrations of the generator force the mass-spring system to oscillate.

Describe and explain the oscillations of the mass-spring system.

[3 marks]

0	6	.	7
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The mass is submerged in a beaker of water while the vibration generator continues to vibrate at a frequency of 1.1 Hz.

Explain why the amplitude of oscillation of the mass is reduced.

[2 marks]

Turn over for the next question

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